

INTEGRATED AIRCRAFT-TRAJECTORY CO-DESIGN

Systems-Level Optimization Framework & End-to-End Implementation Guide

1. PURPOSE & CORE ENGINEERING GOAL

Integrated Aircraft-Trajectory Co-Design mathematically unifies vehicle sizing (wing area, engine size) and operational trajectory (altitude, speed profiles) into a single, coupled optimization problem. Traditional design separates these steps, resulting in performance losses because static vehicle sizing cannot adapt to dynamic operational tradeoffs.

Traditional Sequential Workflow (Decoupled)

- Sizing Phase:** Optimize wing geometry (S, AR) strictly for peak cruise aerodynamic efficiency (L/D).
- Operations Phase:** Find the optimal climb path for the fixed airframe.
- Flaw:** Cruise-optimized wing may be undersized for steep climbs, forcing oversized, heavy engines.

Integrated Co-Design Workflow (Coupled)

- Joint Optimization:** Solves geometry and flight path simultaneously via direct collocation.
- Synergy:** Accepts a slightly less efficient cruise wing if it yields high lift during climb, enabling a lighter engine.
- Result:** Minimizes total mission fuel burn/energy across the entire flight envelope.

2. END-TO-END PROJECT EXECUTION ROADMAP

P1 Requirements & Scope: Define mission Top-Level Aircraft Requirements (TLARs): payload mass ($m_{payload}$), target range (R), and objective function (minimize initial mass m_{total} or total fuel consumed m_{fuel}).

P2 Mathematical Coupling: Define physical interactions across domains:

- Structures & Mass:** $m_{wing} = f(S, AR) \rightarrow$ Total mass $m[i] = m_{payload} + m_{struct} + m_{fuel}[i]$.
- Aerodynamics:** Dynamic lift/drag coefficients: $C_L[i] = 2 m[i] g / (\rho V^2 S)$, $C_D[i] = C_{D0} + C_L^2 / (\pi AR e)$.
- Flight Mechanics:** Collocation dynamics: $dh/dt = V \sin(\gamma)$, $dV/dt = (T - D)/m - g \sin(\gamma)$.

P3 AeroSandbox / CasADi Implementation: Initialize `asb.Opti()`. Declare static decision variables (S, AR) and discretize trajectory variables (h, V, m_{fuel}, T) across $N = 50$ grid points. Apply implicit trapezoidal integration for state transitions and enforce boundary conditions ($h[0]=0, h[N]=10,000m$).

P4 Solving & Trade Studies: Solve nonlinear NLP via IPOPT. Benchmark co-design outputs against the sequential design baseline to quantify total mission fuel reduction. Run sensitivity trade studies on wing aspect ratio vs. mission distance.

3. PORTFOLIO & RESUME PRESENTATION STRATEGY

- Multidisciplinary Aircraft Co-Design Tool (Python / AeroSandbox):** Developed an MDO framework coupling wing planform geometry (S, AR) with dynamic climb profiles via direct collocation in CasADi and AeroSandbox.
- Quantitative Optimization Impact:** Demonstrated a **12% reduction in total mission fuel burn** compared to traditional sequential aircraft sizing by simultaneously solving $N=50$ discretized trajectory states and structural-aerodynamic trade constraints.